Microarchitectural Attacks and Defenses

Assignment 1 - Flush+Reload

Files created:

- flush_reload.c / flush_reload.h
- monitor.c
- fr_trace.c
- gpg_attack.c
- victimshim.c
- Makefile
- gnupg.txt
- trace.out

Implementation summary:

Task 1: map() uses mmap with PROT_READ and MAP_PRIVATE.

Task 2: flush_reload() uses clflush + rdtsc timing.

Task 3: monitor() samples a target with period and maxwait.

Task 4: monitor binary prints detection timestamps.

Task 5: fr_trace tracks two offsets and writes trace.out.

Task 6: gpg_attack.c monitors the trigger and samples madgpg.

Tools used:

- GCC / Makefile build
- GDB and shell debugging
- Linux perf / timing checks
- ChatGPT for code review and debugging support